

Appendix: Table for Two: Human–Robot Co-Adaptation in Month-Long In-Home Mealtime Assistance

Anonymous Author(s)

A WEB INTERFACE

The community researcher (CR) operates the robot through a Vue.js [9] web application on her iPad, which exchanges JSON messages with the robot over rosbridge [3]: the application publishes her taps as messages to the robot, and the robot moves the application to a page by publishing that page’s name. Every message the application sent to the robot over the deployment is logged with its timestamp, and the counts below are taken from these logs; the base joystick and the settings overlay use channels of their own. Figure N2 shows the pages as they appeared at meal 20, from the screen recording of her iPad; Fig. N1 shows the pages that meal 20 did not reach, from the recordings of other meals. Table A1 lists the pages, what she can do on each, and what it sends to the robot. The screenshots are collected at the end of this document, in Appendix N. Her name, which the header shows on every page, is erased from all screenshots.

Elements shared across pages. A header carries a status line describing what the robot expects of her, a *Robot Arm Control* button that opens arm teleoperation, and, on every page but the context and preference pages, a settings gear that opens the settings overlay; while the robot navigates or waits for her to choose a task, a *Robot Base Control* button opens base teleoperation as well. While a skill executes, a strip under the header shows the previous, current, and next skill of the plan, and the page shows a language description of what the robot is doing (Fig. N2p). A wired button at her iPad, read through the iPad’s audio input until meal 20 and through the robot’s computer from meal 21 (Appendix F), can request a bite, a sip, or a wipe in place of the open-mouth trigger.

Start and end of a meal. The home page offers a single *Start Meal* button (Fig. N2a). On the context page, she then chooses the meal, the setting, and the time of day (b, c), which form the context the predictor conditions on (Appendix B). After the plate is in the sink, the end-of-meal survey presents the 11 daily items and the open question of Appendix K one per screen (t), and a closing page confirms that her answers were saved (u).

Preference pages. Each predicted dimension appears on its own page, at the point in the meal where it takes effect, with the predicted value preselected and a 30-second countdown after which it is accepted (Fig. N2d). Three dimensions are asked before the robot leaves for the refrigerator (robot speed and the two confirmation behaviors for navigation and manipulation), one after the plate leaves the refrigerator (microwave time), and the remaining 17 at the table once she presses *Yes, I’m ready* (o). When she selects a different value (e), the page sends it to the robot, which fixes it and re-predicts every dimension still open (f); the pages that follow show the re-predicted values (g). The bite ordering, the one free-text dimension, offers the predicted sentence or a field she can type or dictate into. The settings overlay (Fig. N1a) lists every dimension answered so far in the meal and lets her change

at once any that has not yet taken effect, while those set during meal preparation are shown but locked; she used it 20 times over the month. The nine environment parameters have no preference page: a plate color is set on the color-correction page and a parking offset with the base joystick; both are described below.

Confirmation pages. A confirmation page precedes each step that depends on perception or ungrasps the plate. The detection page serves five detections: door handles, found by GroundingDINO and located on the depth image, and shown with the hinge and the direction in which the door opens (Fig. N2h); the plate attachment, found by color and shown as the pixels that matched (i); the microwave’s start button, returned by Molmo-Point as a point (j); and the placement points on the table (k) and in the sink (l). She taps *Looks Correct* or *Redo*, and for the plate attachment *Correct Color*, which opens the color-correction page (Fig. N1b): she taps the attachment in the camera image, adjusts the matched color range with a slider, reruns the detection until it is right, and confirms, which records the color for that location. The same page opens on its own when the robot cannot find the attachment at all. Before each release of the plate in the microwave, on the table, and in the sink, a release page asks her to confirm (Fig. N2m). On arrival at a task location, a parking page asks whether the base is where she wants it; *Adjust Position* opens the base joystick, and the adjustment she makes is recorded as that location’s parking offset (Fig. N1c). The countdown on each of these pages is itself a personalization dimension: *Confirm manipulation* sets it for the detection and release pages, and *Confirm navigation arrival* for the parking page, each to one of skip, a 15-, 30-, or 60-second countdown, or waiting for her. Skip removes the page from the meal. Over the month, she confirmed 160 detections, 8 of them by letting the countdown run out, asked for 12 re-detections and 12 color corrections, and confirmed 58 plate releases, 1 by countdown; the color-correction page, opened by her or by a failed detection, recorded the 19 plate-color corrections of Section 5.3 of the main paper.

Feeding pages. Task selection offers the three tasks (*Take a Bite*, *Take a Sip*, and *Wipe Mouth*), a *Personalization* menu, and *Finish Feeding* (Fig. N2n). The bite-selection page shows the plate with each detected item boxed and numbered, the bite the robot proposes, the other foods she can switch to, and the dip (q); she can accept, tap another box, or swap to another item (r), and the proposed bite is picked up when the countdown set by *Show bite-selection page* runs out. If no bite is detected, or the robot cannot separate the item she wants, the *Trouble picking this up?* button opens the manual keypoint mode: she chooses a skill and taps one or two points on the full camera image (Fig. N1d, e), which the robot uses instead of the detection; she did so 25 times over 11 meals. After each bite or sip, the next-task page repeats the task selection with a countdown on the previous task, set by *Next-task countdown* (Fig. N1f). While the robot waits for her to request a bite,

Table A1: The pages of the web interface. *Panel* gives the page’s screenshot in Fig. N2 or N1; *Sends* gives what the message each page publishes to the robot on `/webapp_to_robot` carries, with fields in roman and literal values in code. The FEAST personalization pages reached from *Personalization* (language requests and gestures [6]) are omitted: she sent no request from them during the deployment.

PAGE	PANEL	SENDS	CONTENTS AND CONTROLS
Start and end of a meal (4)			
Home	N2a	state home, status start_meal	<i>Start Meal</i>
Meal context	N2b, c	meal, setting, time of day	Meal, setting, and time of day, one per step
End-of-meal survey	N2t	item, value	Eleven 7-point items and one open question, one per screen
Closing	N2u	—	Confirms that her answers were saved
Preference pages (2)			
Preference page	N2d–g	dimension, value, tap or countdown	One dimension, predicted value preselected, 30 s countdown; the bite ordering takes free text
Settings overlay	N1a	dimension, value	Every dimension answered so far this meal, editable at once; opened from the gear on every page but the context and preference pages
Confirmation pages (6)			
Detection	N2h–l	detection type, confirm, redo, or correct_color	Handle, plate attachment, microwave button, table, or sink placement drawn on the camera image; <i>Looks Correct, Redo, Correct Color</i>
Color correction	N1b	color, range, rerun or confirm	Tap the plate attachment, adjust the color range, <i>Rerun, Confirm</i>
Plate release	N2m	location, confirm	Before the plate is let go in the microwave, on the table, or in the sink; <i>Continue</i>
Parking	N1c	ok or adjust	On arrival at a task location; <i>Position OK</i> or <i>Adjust Position</i> , which opens the base joystick
Ready to eat	N2o	confirm	Plate on the table; <i>Yes, I’m ready</i> , with no countdown
Pickup check	—	confirm or cancel	After a bite, the drink, or the wiper is picked up; <i>Continue</i> , and for a bite <i>Retry</i>
Feeding pages (5)			
Task selection	N2n	task	<i>Take a Bite, Take a Sip, Wipe Mouth, Personalization, Finish Feeding</i>
Next task	N1f	task, tap or countdown	Task selection after a bite or sip, with a countdown on the previous task
Bite selection	N2q, r	food item, detection, dip	Plate with numbered detections, proposed bite, food items to switch to, dip; <i>Pickup Bite</i> with countdown; <i>Trouble picking this up?</i>
Manual keypoints	N1d, e	skill, image points	Choose a skill, then tap one or two points on the full camera image
Executing	N2p, s	—	Previous, current, and next skill; a description of what the robot is doing; the trigger it is waiting for
Teleoperation pages (3)			
Arm	N1g	command, step size, done	Task-space moves and rotations at three step sizes, joint-space moves, gripper, retract, <i>Done</i>
After teleoperation	N1h	redo or next	<i>Move the Base, Redo Skill, Next Skill</i>
Base	N1i	velocity on <code>/cmd_vel_teleop</code>	Joystick, drag to drive and release to stop; <i>Resume</i> or <i>Done</i> during a navigation skill

sip, or wipe, the executing page says which trigger it is waiting for (Fig. N2s). The *Personalization* menu retains FEAST’s language and gesture personalization [6]; she sent no request from it during the deployment, and changed every preference on the preference pages or in the settings overlay instead.

Teleoperation pages. On the arm page (Fig. N1g), she moves the gripper in task space, up, down, left, right, reaching out and pulling back, tilting, turning, and rolling, at a fine, medium, or coarse step, opens and closes the gripper, and runs the predefined retract motion; a joint-space tab moves one joint at a time, though she never used it. When she presses *Done*, a page asks what the robot should do next: *Move the Base, Redo Skill*, or *Next Skill* (h). The base page (Fig. N1i) is a joystick she drags to drive the base and releases to stop; during a navigation skill it offers *Resume* to hand the drive back to the planner, and *Done* to end it where she stopped. Appendix I counts how she used both.

Order of pages over a meal. Table A2 gives the order in which the pages appeared at meal 20, a social meal with a friend, reconstructed from the application’s message log and the robot’s event log and checked against the screen recording. Meal 20 had the parking pages set to skip and reached neither teleoperation page; where a meal differs, the pages above appear at the step they belong to.

B PREFERENCE PREDICTION

This appendix formally states the prediction problem summarized in the main paper’s Section 4, describes how the predictor deployed in the home realizes it, and gives the offline loop used to score predictors against the deployment logs.

B.1 Personalization Dimensions

Table B1 lists the 30 personalization dimensions with their options and the point in the meal at which each is asked. Every categorical dimension is predicted from its option list; the bite ordering is predicted as one sentence grounded in the foods on the plate, such as which item comes first and which dip goes with which item. The environment parameters are continuous. A plate color is a hue, saturation, and value with a matching range, set when she corrects the color on the color-correction page (Appendix A); a parking offset is a correction to the base’s recorded waypoint, set when she drives the base on arrival. At meal 1, the plate colors began from a single calibrated value and the parking offsets from zero. The dining table and the movable table began as one location; the system began choosing the table from the setting at meal 4 (Appendix F).

B.2 Problem Formulation

We consider a single user u interacting with the robot over a sequence of meals $t = 1, \dots, T$. The user is characterized by fixed physical functioning abilities U_u^{phys} and by a slowly evolving

Table A2: The pages of meal 20 in order. *Min* counts minutes from *Start Meal*; *Detail* gives the skill or detection the page belongs to, or what happened on it. The feeding loop repeats task selection, bite selection, and execution for each of 15 bites and 3 sips.

MIN	PAGE	DETAIL
Start of the meal		
0.0	Home	<i>Start Meal</i>
0.3	Meal context	meal, setting, time of day
1.2	Preference pages	1–3 of 3: robot speed, confirm navigation arrival, confirm manipulation
Preparation		
1.7	Detection	refrigerator handle
2.9	Detection	plate attachment, refrigerator
4.3	Preference page	1 of 1: microwave time
5.2	Detection	microwave handle
6.6	Plate release	microwave
7.8	Detection	microwave button
8.7	Detection	microwave handle, reopening
9.8	Detection	plate attachment, microwave
14.0	Detection	placement point, table
14.5	Plate release	table
15.2	Ready to eat	<i>Yes, I'm ready</i>
Feeding		
15.2	Preference pages	1–17 of 17; one correction, one re-prediction
17.9	Task selection	<i>Take a Bite</i>
18.5	Bite selection	she swaps the proposed item
18.8	Executing	waiting for her button press
21.5	Next task, bite selection, executing, repeated until 51.1	
53.6	Task selection	<i>Wipe Mouth</i>
54.7	Task selection	<i>Finish Feeding</i>
Clearing		
55.6	Detection	plate attachment, table
56.2	Navigation to the sink stops; the application reloads to the home page and she finishes feeding again	
63.6	Detection	placement point, sink
63.7	Plate release	sink
End of the meal		
63.8	End-of-meal survey	—
72.7	Closing	—

latent state $U_{u,t}^{\text{pref}}$ governing how preferences are formed. The latter is intrinsic to the user, is not directly observable by the robot, and may drift over the horizon of a deployment.

At each meal t , the robot observes an exogenous mealtime context X_t : the meal, the dining context (personal, social, or watching television) and the time of day. The user additionally carries a transient latent state Y_t , varying from meal to meal, that captures factors such as fatigue, hurriedness, or a willingness to try something different. Neither $U_{u,t}^{\text{pref}}$ nor Y_t is observable.

For each meal, the user holds a preference vector

$$P_t = (p_t^{(1)}, \dots, p_t^{(M)}) \in \mathcal{P}, \quad \mathcal{P} := \prod_{i=1}^M \mathcal{P}^{(i)},$$

where $\mathcal{P}^{(i)}$ is the feasible set of dimension i — a finite set of options for the categorical dimensions, a bounded continuum for the plate-attachment colors and parking-pose offsets, and free text for the bite ordering. In this deployment, $M = 30$.

We assume that preference vectors are generated by a shared structural mechanism

$$P_t \sim f(U_u^{\text{phys}}, U_{u,t}^{\text{pref}}, X_t, Y_t),$$

where f defines a joint distribution over $\mathcal{P}$. That the distribution is joint rather than a product of per-dimension marginals is the substantive assumption of this work: the components of P_t are dependent because they are generated from shared latent quantities, which is what makes a correction on one dimension informative about the others.

Interactive prediction. At the beginning of meal t , the robot produces an initial prediction $\hat{P}_t^{(0)}$. The user then confirms or corrects dimensions one at a time, each correction revealing the ground-truth value of a single dimension. After $m \geq 0$ corrections, let $S_{t,m} \subseteq \{1, \dots, M\}$ denote the set of revealed indices with $|S_{t,m}| = m$, let $P_{t,S_{t,m}} = \{p_t^{(i)} : i \in S_{t,m}\}$ denote their ground-truth values, and let $\bar{S}_{t,m} = \{1, \dots, M\} \setminus S_{t,m}$ be the dimensions still open. A dimension the user confirms is likewise fixed for the remainder of the meal and is supplied to the predictor, so $S_{t,m}$ indexes every dimension whose value has been settled, by confirmation or by correction, while m counts the corrections alone. The robot conditions on the interaction history $\mathcal{H}_{t-1}$ and on its own earlier predictions this meal, and outputs

$$\hat{P}_t^{(m)} = g_\theta(U_u^{\text{phys}}, X_t, \mathcal{H}_{t-1}, S_{t,m}, P_{t,S_{t,m}}, \hat{P}_t^{(0:m-1)}).$$

The predictor g_θ never observes $U_{u,t}^{\text{pref}}$ or Y_t directly, and must reason about them through the history and the corrections so far.

Loss and objective. Corrected dimensions are fixed, so prediction reduces to inferring the remaining components $P_{t,\bar{S}_{t,m}}$. With ℓ_i a per-dimension loss and $w_i \geq 0$ optional importance weights,

$$\mathcal{L}_{t,m}(P_t, \hat{P}_t^{(m)}) = \sum_{i \in \bar{S}_{t,m}} w_i \ell_i(p_t^{(i)}, \hat{P}_t^{(m,i)}),$$

and the learning objective is to minimize its expectation over the latent states and the stochasticity of f .

B.3 Realization

The predictor is a single call to a large language model, which returns the full bundle. The interaction history is carried verbatim: $\mathcal{H}_{t-1} = \{h_1, \dots, h_{t-1}\}$, where each episode $h_\tau = (\tau, X_\tau, P_\tau^*)$ records the context of meal τ and the bundle finally accepted in it, serialized as text and supplied in chronological order. No summarization, retrieval, or windowing is applied — the model conditions on the complete record. Alongside it, the call receives a description of U_u^{phys} , the current context X_t with the food on the plate, the dimensions confirmed or corrected so far this meal, and the model's own earlier predictions of the meal, so that it can revise its reading of Y_t where a correction has contradicted one of them. Predictions come from Claude Opus 4.8 at medium reasoning effort.

B.4 Offline Evaluation Loop

The deployed joint predictor cost the CR 57 corrections over the 22 meals, as counted during the deployment. To compare this observed workload with alternative predictors, we replay each meal under a common offline protocol (Algorithm B1). Each

Table B1: The 30 personalization dimensions and their options. The 21 meal preferences are grouped by what they govern: when the robot checks with her, the signals between them, settings fitted to her functional abilities, the bite itself, how the robot moves, and heating; the main paper’s Section 4 counts the fit to her abilities and the bite together, as five dimensions on the bite. Within a group, dimensions are listed in the order they take effect in a meal. *Asked* is the point in the meal at which the dimension’s preference page appears: before the robot leaves for the refrigerator (*start*), after the plate leaves the refrigerator (*fridge*), or at the table once she is ready to eat (*table*). The environment parameters have no preference page and are set at each location, on pickup or on arrival. Dimensions marked * held the same value at all 22 meals.

DIMENSION	ASKED	OPTIONS
MEAL PREFERENCES (21)		
Checks with her (5)		
Confirm navigation arrival	start	skip · countdown 15, 30, or 60 s · wait for me
Confirm manipulation	start	skip · countdown 15, 30, or 60 s · wait for me
Show bite-selection page*	table	countdown 15, 30, or 60 s · wait for me
Confirm feeding pickup*	table	skip · countdown 15, 30, or 60 s · wait for me
Next-task countdown	table	countdown 15, 30, or 60 s · wait for me
Signals (8)		
Robot’s ready-to-feed cue	table	speech · LED · speech and LED · no cue
Completion signal	table	speech · LED · speech and LED · no cue
Bite, drink, and wipe trigger (3)	table	open mouth · button · proceed automatically after a pause
Done detection (bite)	table	perception (force/torque) · button · proceed automatically after a pause
Done detection (drink, wipe) (2)	table	perception (head nod) · button · proceed automatically after a pause
Fit to her abilities (3)		
Skewering axis*	table	parallel · perpendicular (to the item’s major axis)
Transfer mode*	table	outside mouth · inside mouth
Outside-mouth distance*	table	near · medium · far · not applicable
Bite (2)		
Bite dipping	table	do not dip · less · more
Bite ordering	table	free text, such as the order of items and which dip goes with which item
Motion (2)		
Robot speed*	start	slow · medium · fast
Retract between bites	table	yes · no
Heating (1)		
Microwave time	fridge	no microwave · 30 s · 1 min · 2 min
ENVIRONMENT PARAMETERS (9)		
Plate color (4)		
Refrigerator, microwave, dining table, movable table	on pickup	hue, saturation, value, and a matching range; set on the color-correction page
Parking (5)		
Refrigerator, microwave, dining table, movable table, sink	on arrival	$\Delta x, \Delta y$ within ± 0.5 m and $\Delta \theta$ within $\pm 45^\circ$ of the recorded waypoint; set by driving the base

alternative predictor is shown the meal’s context and predicts the bundle; any open dimension that disagrees with the bundle the CR finally accepted is eligible to be corrected; one such dimension is revealed and the predictor runs again; the meal ends when no open dimension disagrees.

The one respect in which this departs from the deployment is the choice of which dimension to reveal. In the home, the CR corrects dimensions in the order the system surfaces them, and her counterfactual order under a different predictor is unknowable, so the loop reveals a uniformly random disagreeing dimension and results are averaged over reveal orders.

A correction is counted whenever the CR sets a value the system did not already hold. This includes the three occasions on which she revised a dimension she had already settled earlier in the same meal; these reflect her own reconsideration rather than a prediction failure, and are counted against both LLM arms alike. The

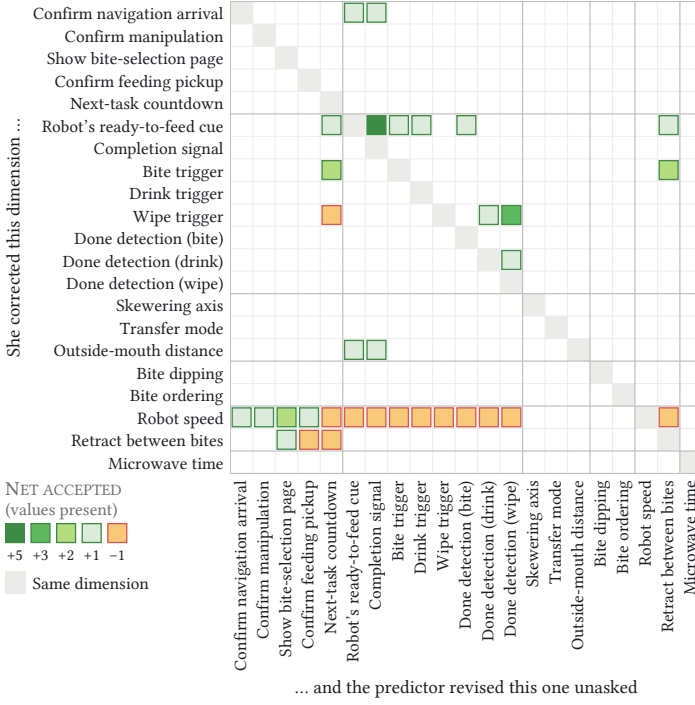
count-based arms are scored under this same protocol, each predicting day t from days 1 to $t - 1$.

Predictors are compared on two quantities: *accuracy-after- m* , the prediction accuracy on the open dimensions $P_{t, S_{t,m}}$ as a function of the m corrections received; and *corrections-to-stop* m_t^* , the number of corrections needed before the prediction matches P_t^* on every open dimension. Summed over the deployment, $\sum_t m_t^*$ is the total correction workload in Section 5 of the main paper.

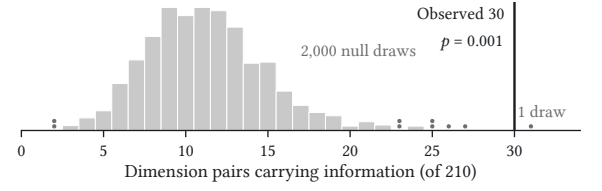
B.5 Correlation Evidence

The evidence behind the correlated-bundle hypothesis is in Fig. B1: which dimensions propagated to which (a), that the dependence exceeds chance (b), and the per-meal correction counts underlying the cumulative comparison in the main paper (c). All three panels cover the same 21 categorical and free-text dimensions; the

(a) One correction moves others



(b) Dependence vs. chance



(c) Every arm, meal by meal

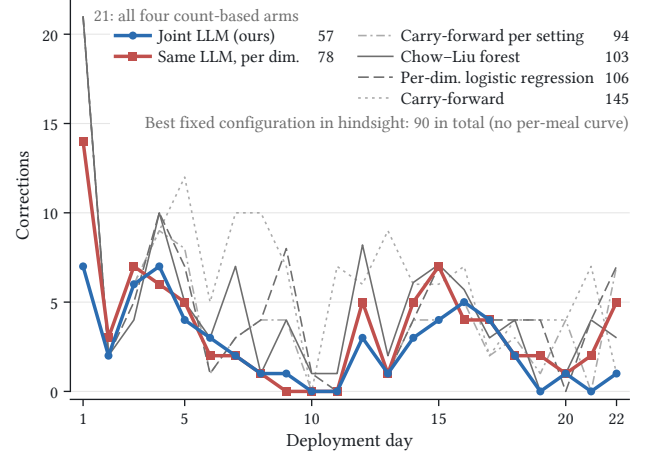


Figure B1: The CR’s preferences are correlated, and exploiting them lowers her correction workload. (a) Rows are dimensions she corrected, columns those the predictor then revised unasked, both ordered and grouped as in Table B1; shade is accepted revisions minus overwrites of already-correct values, green where propagation net helped, amber where it hurt. (b) Count of informative dimension pairs under 2,000 circular shifts (gray) against the observed count. (c) Each arm’s corrections at every meal, underlying the cumulative totals in the main paper; the key gives each arm’s deployment total. *Same LLM, per dim.* predicts each dimension on its own, as CBTL [8] does; *Carry-forward per setting* is FEAST’s rule [6] (Appendix C).

Algorithm B1: Offline evaluation at meal t

Input: U_u^{phys} , history $\mathcal{H}_{t-1}$, context X_t , accepted bundle P_t^*
Output: corrections m_t^*

- 1 $m \leftarrow 0, S_{t,0} \leftarrow \emptyset;$
- 2 **repeat**
- 3 $\hat{P}_t^{(m)} \leftarrow g_\theta(U_u^{\text{phys}}, X_t, \mathcal{H}_{t-1}, S_{t,m}, P_{t,S_{t,m}}, \hat{P}_t^{(0:m-1)});$
- 4 **if** $\exists i \in \bar{S}_{t,m} : \hat{p}_t^{(m,i)} \neq p_t^{*(i)}$ **then**
- 5 $j \sim \text{Uniform}(\{i \in \bar{S}_{t,m} : \hat{p}_t^{(m,i)} \neq p_t^{*(i)}\});$
- 6 $S_{t,m+1} \leftarrow S_{t,m} \cup \{j\}, P_{t,S_{t,m+1}} \leftarrow P_{t,S_{t,m}} \cup \{p_t^{*(j)}\};$
- 7 $m \leftarrow m + 1;$
- 8 **until** $\hat{P}_t^{(m)} = P_t^*$ on all $i \in \bar{S}_{t,m};$
- 9 $m_t^* \leftarrow m;$
- 10 $\mathcal{H}_t \leftarrow \mathcal{H}_{t-1} \cup \{h_t\}, h_t = (t, X_t, P_t^*);$

nine environment parameters (plate colors and parking offsets) are excluded as perceptual and geometric values, not meal preferences.

Panel (a) reads the correlation that the predictor actually used directly from the logs: a row is a dimension the CR corrected, either on its preference page or mid-meal on the settings page, and a col-

umn is a dimension the predictor then revised on its own, without being asked about it. Each cell is outlined in its arm’s darkest step, which is what separates the two pale tints from one another. The effect is real but uneven: correcting the interaction cue for one task carried most reliably to the same cue for the others (*Robot’s ready-to-feed cue* $\rightarrow$ *Completion signal*, five accepted revisions and none wrong), whereas a single mid-meal speed edit on meal 4 fanned out to 12 dimensions and mostly overwrote correct values (the *Robot speed* row).

Test of correlation. Panel (b) tests whether the dimensions carry information about one another in the values she accepted. Six of the 21 meal preferences held the same value at every meal (Table B1), leaving 15 that varied and $15 \times 14 = 210$ ordered pairs. For each ordered pair, we predict the second dimension from the first by a lookup table fitted on every other meal, leaving out the meal being predicted, and compare its accuracy with that of the second dimension’s own most common value, fitted the same way. In 30 of the 210 pairs, the first dimension predicts the second better. The null circularly shifts each dimension’s 22-meal history by an independent random offset, which preserves how sticky each preference is over the deployment and destroys only the alignment between dimensions; over 2,000 shifts it yields 11.25 such pairs on average, and one shift reaches 30 or more ($p = 0.001$).

Propagation accounting. Every re-prediction over the deployment is logged with the values it changed. Scoring each changed meal preference against the value she held at the end of that meal, the predictor revised 62 values without being asked: 37 revisions turned a wrong value right, 22 turned a right value wrong, and 3 replaced one wrong value with another, a net gain of 15. They came from 23 re-predictions, which changed 2.7 values on average; the largest, the mid-meal speed edit of meal 4, changed 12, most of them values that were already correct.

Amplification. The amplification reported in Section 5.3 of the main paper is computed over the corrections she made on the preference pages, where each correction triggers a re-prediction of the dimensions still open: it is the number of dimensions, meal preferences and environment parameters alike, that these re-predictions changed, divided by the number of such corrections. Over meals 1 to 6, 15 corrections changed 21 dimensions (1.4); over meals 12 to 18, 19 changed 24 (1.3); over meals 19 to 22, 2 changed 7 (3.5). The one such correction of meals 7 to 11 changed none.

In panel (c), corrections are counted as individual events, so the panel agrees meal by meal with the timeline table in the main paper; the only difference is that the plate-color channel plotted there has no counterpart here. Meal 1 costs every count-based arm all 21 dimensions, having no history to predict from, whereas both LLM arms begin from a physically grounded prior; this is what sets the panel’s vertical range.

C BASELINES

Every alternative in Fig. 3 of the main paper predicts meal t from meals 1 to $t - 1$ and is scored under the replay protocol of Appendix B.4. We chose one representative of each family a designer could have deployed before LLM-based prediction, and one oracle as a ceiling.

Same LLM, per dimension. The same model and reasoning budget as the joint predictor (Claude Opus 4.8, medium reasoning), but one call per open dimension, each shown only that dimension’s own history of (day, meal, setting, time, value) and her physical-functioning description. It never sees another dimension’s value or her corrections that meal, and is not re-run after a correction, so its errors are its corrections. This is the structure of CBTL [8], which learns each parameter independently; against it the joint predictor isolates the value of correlation alone.

Carry-forward. For every dimension, repeats her last accepted value: the intuitive strawman, and the model an “auto-continue” setting implements. It costs 145.

Carry-forward per setting. Repeats the last value she accepted in the same setting (personal, social, or television), falling back to plain carry-forward until a setting has been seen. This is what FEAST [6] does: it stores one bundle per context and reloads it. It is the strongest count-based arm at 94, which shows that most of plain carry-forward’s cost is dragging one setting’s bundle into another, not drift.

Per-dimension logistic regression. A multinomial softmax per dimension over one-hot setting, time of day, and meal name, fitted by gradient descent (400 steps, learning rate 0.5, L_2 weight 0.1) on meals 1 to $t - 1$. With no history, a dimension is counted wrong; with a single observed value that value is predicted. This

is the full-information reduction of the contextual-bandit view of preference learning. Adding the meal name never helps (106 to 116 across encodings); it costs 106.

Chow-Liu forest. The textbook joint categorical model for small n [2]. Pairwise mutual information is computed from counts with add-0.5 smoothing over meals 1 to $t - 1$; an edge is kept if $2n$ MI exceeds the BIC penalty $\ln(n) (|A| - 1)(|B| - 1)$, and Kruskal’s algorithm on the kept edges gives a forest. Prediction clamps the dimensions corrected so far and runs exact sum-product in each component, taking the argmax belief; a revealed value never seen before falls back to the marginal. With fewer than four meals of history, it predicts the per-dimension mode. Reveal order is random; the total, 103, averages 10 orders.

Best fixed configuration in hindsight. The single constant bundle minimizing total corrections over all 22 meals, which is the per-dimension mode, chosen with knowledge of the whole deployment. It costs 90 and is a ceiling on any non-adaptive predictor, not a baseline. Guessing uniformly from each dimension’s option list would cost roughly 324 corrections.

Two scoring rules favor the count-based arms. Her three same-meal revisions (meals 3, 5, and 6), in which she changed a dimension she had already settled, are charged to both LLM arms and not to the count-based ones. And a count-based arm cannot produce a value it has never observed; 44 of the 462 scored slots hold one. The joint predictor faces the same cold start on meal 1, where a count-based method has no history and misses all 21 dimensions, and costs 7 corrections there, the hindsight bundle’s cost that day.

D PREPARATION PHASE

We spent five weeks, from June 11 to July 16, 2026, developing and testing the system in the CR’s home before the first meal of the study. Table D1 gives a day-by-day account, reconstructed from the dated records the work left: the system’s code history, the maps built of her home, the logs of every run of the robot, and the notes we kept as we worked, and from a team member’s journal and calendar of the period. We worked in her home on 29 of the 36 days; on the other seven we traveled, or she was unavailable and we worked elsewhere.

From skills to a complete meal. The first two weeks brought the manipulation skills to her appliances and the interface around them: the refrigerator, microwave, and sink skills worked in her kitchen by June 18, the first map of her home was built on June 20, and the first run of the complete pipeline, from refrigerator to sink, followed on June 21. The hardware needed changes of its own: the microwave-safe plate attachment did not dock in its holder until we sanded the holder’s mating part (June 17), and the feeding utensil’s holder broke during an integrated run and was reprinted (June 25). From June 25, members of the research team ate stand-in meals with the robot most days, which exercised the pipeline end to end and surfaced failures before the CR used it.

Navigation. Most of the remaining time went to navigation, for which her home leaves little margin. We built eight maps as we refined the system, and three failures shaped the configuration that ran during the study. First, the table’s outline, as the lidars see it, is nearly symmetric, so localization could flip by 180° beside it (July 6). Second, the robot’s odometry initially came from the ZED

Table D1: The preparation phase, day by day. Every date from June 11 to July 16, 2026, on which the records show work on the system or a change of plan; the team worked in her home on 29 of the 36 days. *Stand-in* meals were eaten by research team members, not the CR.

DATE	WORK	DATE	WORK
Week 1 The robot in her kitchen		Week 3 Stand-in meals	
Jun 11	Travel to her town; no work on the robot	Jun 25	An integrated run from the refrigerator to the table; the feeding utensil's holder broke and was reprinted
Jun 12	Her home unavailable: design of the study and of the interface for taking over a manipulation skill	Jun 26	Stand-in meals, begun the day before; emergency-stop dropouts traced to the wireless channel and fixed; resuming a skill after teleoperation
Jun 13	First day in her home: skirts on both tables, a cover on the refrigerator door, and tape marking each parking location on the floor; opening her refrigerator; teleoperation of the arm and base from the web interface as a fallback for failed skills	Jun 27	Third map; collision thresholds set per skill
Jun 14	Microwave manipulation; detection confirmation pages in the interface	Jun 28–30	Stand-in meals on most days; sensor diagnostics; the door of her microwave opened wider
Jun 15	Microwave manipulation and the web interface; testing ended early because she had a commitment	Jul 1	Fourth map; parking offsets learned from her adjustments on arrival
Jun 16	The new microwave-safe plate attachment found to fit its holder poorly; remote emergency stop built	Week 4 Navigation	
Jun 17	Microwave manipulation; the holder's mating part sanded until the attachment docked reliably	Jul 2	Navigation logging; the visual-inertial odometry found to diverge while the robot's base was parked
Week 2 Every appliance skill and the first complete run		Jul 3	Fifth map; prediction of all personalization dimensions; the confirmation pages and their countdowns designed
Jun 18	The refrigerator, microwave, and sink skills work in her kitchen; the microwave-button model served remotely; taking over extended from arm to base; the wired button at her iPad	Jul 6	Localization found to flip by 180° at the table, whose view is nearly symmetric; the plate attachment's color changed from teal to orange
Jun 19	Her home unavailable: design of learning from her corrections	Jul 7	Wheel encoders read back to the robot's computer
Jun 20	The first map of her home; refrigerator manipulation completed and table manipulation begun; navigation's goal-reaching refined and its arrival tolerance logged	Jul 8	Wheel odometry calibrated on the floor of her home; in the evening, a test of the complete pipeline
Jun 21	First run of the complete pipeline, from the refrigerator to the sink, over nine hours; the second map; the color-correction page; resuming an interrupted navigation	Jul 9	Visual-inertial odometry found to drift in direct sunlight with no fault reported; wheel commands made reliable; base speeds calibrated
Jun 22–23	Navigation refined; the base's controller moved to the arm's computer; the remote emergency stop put into use	Jul 10	Base speeds lowered; fusion of wheel odometry and the IMU tested alongside the camera; the team returned home for the weekend
Jun 24	Logging of every meal event; staged preference pages; preference prediction moved to Claude	Week 5 Wheel-and-IMU odometry, the trial meal	
		Jul 13	RealSense firmware downgraded to stabilize its streams; the switch to wheel-and-IMU odometry, validated in a run in which the camera's estimate drifted 5.9 m while localization held; sixth map
		Jul 14	Trial meal with the CR; a fallback when wheel slip turns the base during a scripted drive
		Jul 15	Localization protected from scans occluded by the arm and people at the table; the camera's visual odometry turned off after its software crashed during a run; the seventh and eighth maps, of which the eighth was used from meal 1
		Jul 16	End-of-meal survey; a full stand-in rehearsal of the meal

Mini's visual-inertial odometry, which failed in the home in three ways: it drifted under direct sunlight on the lens while reporting no fault (July 9), crept slowly off course by several meters, and its software crashed four times. We added encoder read-back to the wheels (July 7), calibrated wheel odometry on her floor (July 8), and on July 13 switched to an extended Kalman filter fusing the wheel encoders with the yaw rate of the ZED's IMU; in the run that validated it, the camera's own estimate drifted 5.9 m while localization was unaffected. On July 15, after the camera's software crashed during a run, we turned its visual odometry off and kept only its IMU. Third, while the base was parked at the table on July 14, the arm and the people around it occluded the lidars, and localization jumped by meters; from July 15 the system discards scans too sparse to localize against.

Additions to the home. On our first day in her home, June 13, we made three small additions to her home. We put a floor-length skirt on each table, gingham on the dining table and a plain one on the movable table, so that the lidars see a surface rather than the table's legs. We put a cover on the refrigerator door, so that glare

from sunlight would not corrupt the depth readings the robot uses to find its handle. And we marked each parking location on the floor with tape, as the reference against which we measured how precisely the base parked.

Planning with the CR. We planned the study together with her: the three settings in which she expected to use the robot (eating on her own or while watching television at the movable table, and eating with a friend at the dining table) and the schedule of the meals, set around her availability, which moved the start of the study by about a week.

The trial meal. The CR's only exposure to the complete system before the study was a single trial meal on July 14, three days before meal 1. The robot ran the whole pipeline, from the refrigerator to the sink. The trial did not seed her preferences: it was logged separately from the study, and the predictor's history at meal 1 was empty, so its first predictions drew only on her physical functioning and the meal's context. The one value carried into the study was the plate attachment's color, calibrated during a test that day, from which all plate-color dimensions began.

Table E1: The six settings she ate in. *Label* is the setting she chose on the meal-context page, which the predictor conditions on; the page’s labels were reduced to these three at meal 16. The dining table serves social meals and the movable table all others. Swatches are the setting colors of Fig. 3 in the main paper.

SETTING	LABEL	MEALS
Planned before the study (3)		
 Watching TV	TV	1, 2, 5, 8, 11, 14, 19
 Alone	Personal	3, 6, 15
 With a friend	Social	4, 7, 9, 10, 13, 20
Proposed by her (3)		
 Music on	Personal	12
 Video call	Social	16, 17, 22
 iPad games	Personal	18, 21

E RESEARCHER NOTES AND INTERVENTIONS

Settings. The CR ate in six settings over the month (Table E1). We planned three with her before the study: eating while watching television and eating on her own, both at the movable table, and eating with a friend at the dining table. She proposed the other three as she grew comfortable with the robot: music at a personal meal from meal 12, video calls with her daughter at the dining table from meal 16, and games on her iPad from meal 18. These three are settings she had never eaten in with a human caregiver. She also ate her first morning meal with the robot at meal 15.

The corpus. The attending researcher kept timestamped notes during each meal in the system’s logging interface. They cover all 22 meals; the notes of meals 5 and 20 were written into the file of the preceding meal, and are counted with the meal at which they were taken. By episode, the corpus holds 100 notes and 27 explanations, of which 91 and 20 carry free text. The researcher also marked intervals as interventions during the meal; these marks were one source for the count below, not the count itself. The researcher sometimes began the notes late, as at meals 12 and 16 (“I started taking notes late”), so the counts are totals over the month, not a per-meal series. The CR’s name has been replaced with “the CR” throughout.

Interventions. We identified the 32 interventions by combining the robot’s execution log, the researcher’s in-meal notes, and the video recordings of the meals, counting an episode when the run had to be stopped, or the robot repaired, by someone other than the CR, and attributing each to a cause from the same sources. Tables E3 and E4 list them by meal. By cause, 11 were hardware (gripper fingertips working loose or breaking on meals 4, 9, and 10; cables caught on meals 7, 9, 12, and 15; the depth sensor failing on meal 11; the battery running out on meal 17; the transfer button registering only intermittently on meal 20), 17 software (navigation on meals 7, 10, 13, 20, and 21; food detection on meals 15, 19, and 22; plate detection on meals 19 and 21, the second in a room too dark for it; a code change that broke bite transfer on meal 12; unexpected arm motion on meals 4 and 16; an arm fault that left a service unavailable on meal 17; and single faults in collision checking, mouth-open detection, and the network), and four interface. By skill, bite acquisition and navigation account for six each, bite transfer for five, and the refrigerator for four. Three of the interface

interventions share one cause, an interface bug: a takeover the CR pressed during bite transfer (meals 9, 14, and 17) was not handed to her, and the system stopped responding until the researcher restarted it. This was fixed after meal 17.

Several kinds of episode are excluded from the count. The CR’s own takeovers are low-level teleoperation and are reported with her corrections in Section 5. Nine caregiver tasks she delegated to the researcher, as she would to a human caregiver, are not system failures: reheating food three times, cutting steak, a towel for dripping, removing two pieces of pineapple too hard to chew, moving a food item the plate hid, and moving the plate past a joint limit. Three episodes began with an input error rather than a system fault: the CR choosing *done* instead of *resume* after a failed handle detection (meal 7), and the emergency stop struck by mistake (meals 14 and 16). Twice the researcher returned the iPad to the task-selection page while the robot kept running (meals 15 and 18).

Explanations. 27 episodes record the CR asking the research team something rather than the system: 23 in the first 11 meals, 4 in the last 11, none after meal 15. Four concern navigation or repositioning the base, two of them how to resume after it.

Corrections known to be input errors. At least four corrections in the ledger of Appendix J are logged as misclicks the CR then reverted or continued with: a plate color on meal 3, a setting of a confirmation page on meal 5, an unspecified correction on meal 6, and a base-position confirmation on meal 8. They are counted in the totals of the main paper, the plate color among the 19 plate-color corrections and the rest among the 57 preference corrections.

Survey corrections. Two daily responses were corrected at the CR’s request after she identified mis-entries: safety on meal 8 (3 → 6) and satisfaction on meal 16 (1 → 7). The meal-8 note records the mis-entry contemporaneously (“For safety, she wanted to put 6, but put 3 instead by mistake”). The raw logs retain the original entries.

What she learned each day. Each end-of-meal survey closed with the question “What, if anything, did you learn about the robot today, or how did your interaction with it change?” Table E2 reproduces every answer as she gave it, by dictation or typing on her iPad, with capitalization, spacing, and final punctuation corrected; she left it blank at six meals.

F CHANGES TO THE SYSTEM DURING DEPLOYMENT

As in prior in-home deployments [7], we did not freeze the system: we changed it between meals when a failure revealed the need, and Table F1 lists every change with the first meal it applied to. Because changes were often committed in batches after a meal, we date each one from the code that actually ran: at every start of the robot’s software, the system recorded the version of the code and every edit not yet committed, which gives the exact code that ran at each of the 22 meals. We checked the dates against the robot’s behavior-tree parameters and preference dimensions logged at each meal and against the researcher’s notes. Changes to analysis and study tooling that did not affect the robot are excluded.

Three changes are the ones Section 5.2 of the main paper refers to. The drinking skill became functional at meal 12, when the mug

Table E2: What she learned, meal by meal. Her answer to the question that closed every end-of-meal survey, “What, if anything, did you learn about the robot today, or how did your interaction with it change?” Answers are verbatim but for capitalization, spacing and final punctuation; a bracketed word replaces one misrecognized by dictation. Quotations in the main paper are lightly edited, with her permission. She left the question blank at meals 10, 11 and 14–17.

MEAL	ANSWER	MEAL	ANSWER
	1 I learned that I can change any setting at any time and on any screen.	13	I need to remember to look at the robot hand after pressing the button for it to bring the bite to my mouth.
	2 I am starting to trust the robot’s movements more.		
	3 Today I learned how to control the robots arm myself and restart a skill in case of an error.	14–17	<i>left blank</i>
	4 Today I used the button to cue the robot to bring the food to my mouth. I liked this option for the social setting because I didn’t feel rushed and could carry on a conversation.	18	Split screen works very well for playing games. Sometimes, especially on the screen used to move the robot arm, some options are not showing. I had to use the robot’s voice to [tell] me to open my mouth because I am too busy looking at the screen. When the movable table was accidentally hit, the robot joints hit a limit when it tried to pick the plate up so now I will be extra careful when moving around the movable table.
	5 I noticed the robot predicted that the strawberries would not be microwaved. It predicted that I like 30 seconds in between bites.	19	As the lighting outside changed, the robot struggled identifying the pieces of food on my plate. I was also more careful about not hitting the table, but when I did I was sure to bring attention to it so it could be fixed.
	6 I had to control the robot base today so it would not crash! Once I corrected the path of the robot, it did a complete 360 to acquaint itself with its surroundings.	20	Had trouble with the button signaling to give me a bite. Thinking it is a hardware problem. Once I did not hit the button but it moved towards me. Figured out it was due to the iPad microphone being on and a plastic container loudly opening.
	7 The robot is very sensitive to lights and shadows. I need to be aware if I need to use the continued option or a resume option after moving the robot arm.	21	As I ate, the lighting changed so at the end of my meal when the robot went to pick up my plate it could not distinguish the orange from the rest of the background due to it being too dark in the room. I had to ask for assistance turning on the lights.
	8 I learned that the robot may be very sensitive to my wheelchair moving around it. Also, just a slight move of the base or arm can help complete an action.	22	I found a way to talk and not trigger the robot to feed me a bite when seeing my “open mouth.”
	9 I need to remember to look straight at the robots arm after I press the button for it to advance to my mouth.		
10–11	<i>left blank</i>		
	12 I learned that opening my mouth is not my preferred gesture for face wiping. I chose which bite to take because the robot could not distinguish between the two different types of food.		

moved to a fixed holder on the base from which the robot picks and stows it; none of the drinking skills ran at meals 1 to 11, and each ran at every meal from 12 to 22. The navigation fix applied from meal 14: the gyroscope’s messages carried timestamps that the odometry filter could reject, in which case it estimated the base’s rotation from the wheels alone, which under-report rotation. Which messages it rejected varied from one start of the software to the next, so the failure appeared at some meals and not others; it spoiled the base’s parking at meals 8 and 13. Re-stamping each message with the time it is received removed it. The gripper’s 3D-printed fingertips were repaired during meals 4, 9, and 10 without a restart of the meal.

The remaining changes fall into four kinds. Many addressed a failure that the previous meal exposed, such as the transfer target raised after the fork stopped below her mouth at meal 3, or the collision threshold raised after a false stop at meal 7. A second kind followed her use of the robot: the dining and movable tables became separate locations at meal 4, her first social meal at the dining table, and the button moved to the robot’s computer after it misfired at meal 20. A third kind recalibrated the poses of her furniture and her appliances. The fourth kind, before most meals, added object-detection prompts for the foods of that meal; when a prompt failed during a meal, we corrected it and restarted, which happened at meals 15, 19, and 22.

G MEALS OF THE DEPLOYMENT

Figure G1 shows every meal of the deployment as served, one panel per meal. The CR chose each meal’s food, so the set was not

curated for the robot: it spans dry and crumbly items (hash brown, pancake) through sauced and slippery ones (orange chicken) to soft fruit, and varies in piece size, count, and spacing. It is the range of foods the acquisition stack had to handle in the wild, and is referred to from Section 5 of the main paper.

H PREFERENCE BUNDLE BY MEAL

Where Table J1 records the individual overrides, Table H1 records the result: the complete preference bundle the system held at the end of every meal, one column per meal and one row per dimension. Each column is headed by that meal’s setting and plate, so the configuration can be read against the context that produced it. Cells the CR corrected during the meal are shaded and bold; the rest are values she accepted as predicted.

Read across a row, the table separates dimensions she never revisits from ones that move. Six are constant over the whole deployment: *Skewering axis*, *Transfer mode*, *Outside-mouth distance*, *Confirm feeding pickup*, *Robot speed*, and *Show bite-selection page*. The dimensions that do move cluster rather than drift independently. The done-detection dimensions settle over the first week. The drink and wipe triggers shift once those two tasks enter regular use from meal 12, and shift again in the social meals of meals 16 and 17 — the *Drink trigger* had been button in every social meal and open mouth in every TV or personal meal up to meal 16, and meal 17 is the first social meal where she asks for open mouth instead. The *Next-task countdown* moves the same way, from *wait for me* in the five earlier social meals to a 60-second countdown on meals 16 and 17.

Table E3: The researcher interventions over meals 1–13. *Category* is the cause. *Skill* is the high-level action affected, from the robot’s log. Each description combines the robot’s log, the researcher’s note, and the video recording of the meal. Meals not listed had no intervention.

MEAL	CATEGORY	SKILL	WHAT HAPPENED
3	Software	open_fridge	A network error stopped the system before the refrigerator opened; the researcher restarted it and reran the skill.
4	Hardware	open_fridge	A gripper fingertip worked loose; after the CR’s three retries, the researcher fixed it during her fourth takeover.
	Hardware	pick_utilensil	The fingertip fix did not hold and the utensil could not be picked up; the researcher fitted the spare fingertip.
	Software	acquire_bite	The arm began an unexpected trajectory during bite acquisition; the researcher stopped and relaunched the system.
7	Hardware	pick_plate_from_microwave	A cable that had not been fully unfurled was pulled; the researcher stopped the run and freed it.
	Software	navigate_to_dining_table	Navigation raised transform-extrapolation errors; the researcher restarted it.
	Software	acquire_bite	On the CR’s retry, a collision threshold set too tight stopped the software; the researcher restarted it.
9	Hardware	open_fridge	A fingertip broke; the researcher stopped the system and replaced it.
	Hardware	pick_utilensil	A cable caught on the mouth wiper; the researcher stopped the system and freed it.
	Interface	transfer_utilensil	Due to an interface bug, the robot ignored the CR’s takeover during transfer and stopped responding to the iPad; the researcher restarted it.
10	Hardware	open_fridge	A fingertip worked loose again; the researcher tightened it during the CR’s takeover.
	Software	navigate_to_sink	Navigation failed after the CR had moved; the researcher reset it and relaunched.
11	Hardware	navigate_to_microwave	The depth sensor failed; the researcher recovered it and relaunched.
12	Software	transfer_utilensil	A code change made earlier that day broke bite transfer; the researcher fixed it and relaunched.
	Hardware	pick_wipe	A cable caught and the collision sensor tripped; the researcher stopped the system and freed it.
13	Software	navigate_to_dining_table	Navigation failed twice in succession; the researcher reset it and relaunched.

Table E4: The researcher interventions over meals 14–22. Columns as in Table E3.

MEAL	CATEGORY	SKILL	WHAT HAPPENED
14	Interface	transfer_utilensil	Due to an interface bug, the robot ignored the CR’s takeover during transfer of a bite she had let cool, and stopped responding to the iPad; the researcher restarted it.
15	Software	acquire_bite	The food-detection prompt did not detect the pancake; the researcher stopped the system and revised it.
	Hardware	stow_utilensil	A cable lay in the stowing path; the arm struck it, triggering the emergency stop, and the researcher moved it.
16	Software	transfer_drink	The drink moved down and to her right, not toward her mouth; after the emergency stop, the researcher reset it.
17	Software	open_microwave	The arm jerked and faulted while opening the microwave, and a service it needed became unavailable; the researcher reset the system.
	Interface	transfer_utilensil	The CR missed a bite and took over to retry the transfer; due to an interface bug, the robot ignored the takeover and stopped responding to the iPad; the researcher restarted it.
	Hardware	acquire_bite	The battery ran out and the arm dropped; the researcher recovered it.
18	Software	transfer_drink	Mouth-open detection misfired; the CR redid the sip, but detection then stopped responding and the researcher restarted the system.
	Interface	pick_plate_from_movable_table	The iPad showed no plate detection and the table had moved; the researcher relaunched and reset the setup.
19	Software	acquire_bite	Food detection missed the chicken, labeling it pineapple; the researcher restarted with a revised prompt.
	Software	pick_plate_from_movable_table	Plate detection failed repeatedly despite the CR’s corrections; the researcher stopped the run and relaunched.
20	Hardware	transfer_utilensil	The transfer button registered presses only intermittently; the researcher reconnected it.
	Software	navigate_to_sink	Localization jumped while parked, leaving no feasible path; the researcher stopped the run and relaunched.
21	Software	pick_plate_from_movable_table	The room was too dark for plate detection; the researcher relaunched and, at the CR’s request, turned on the lights.
	Software	navigate_to_sink	Leaving the table, the base reversed and turned; the researcher stopped it and relaunched.
22	Software	acquire_bite	The food-detection prompt read the whole sandwich as one region, so the plate appeared empty; the researcher revised it.

Table F1: Every change to the system during the deployment. Changes to the software on the robot’s main computer and to the hardware, each listed at the first meal it applied to; the software changes run in meal order, followed by the recurring prompt change and then the hardware changes. A change made *during* a meal took effect at that meal; every other change was made between meals.

MEAL	COMPONENT	CHANGE
2	Manipulation	Arm speed presets retuned so joint and Cartesian motions match at each speed; Cartesian trajectories slow to a stop at their last waypoint
	Manipulation	Outside-mouth distances shortened to 5, 7, and 10 cm for near, medium, and far
	Manipulation	Bite-acquisition history kept per meal rather than carried across meals
	Safety	Speed held at <i>slow</i> while the arm picks, stows, or carries a tool or the plate, whatever her speed preference
	Personalization	The predictor also estimates her transient state and receives its own earlier predictions of the meal (Appendix B.3); it retries when a response lacks the full bundle
	Interface	The settings overlay lists dimensions in the order they are asked
4	Navigation	The dining table and the movable table become separate locations, chosen from the setting, each with its own parking offset; a new map of the home with waypoints for the dining table and the kitchen exit
	Navigation	The map correction frozen while the base is parked, to keep occluded scans from moving it (removed before meal 5)
	Perception	Transfer waits for a valid head-pose reading instead of failing when none arrives
	Manipulation	Outside-mouth transfer target raised by 3 cm, after the transfer at meal 3 stopped below her mouth
	Safety	The takeover button is honored during bite acquisition
	Interface	The language descriptions of the robot’s actions generated by Claude Opus 4.8 in place of Claude Haiku 4.5
5	Navigation	The route out of the kitchen depends on where the base leaves from; waypoints at the microwave, sink, and movable table re-recorded
	Perception	Plate color and plate height kept separate for the dining and the movable table
	Perception	Head-pose readings accepted up to 1.5 m instead of 1 m; a 1 s pause after picking a tool
	Manipulation	Gripper opening and closing setpoints recalibrated, after a fingertip broke at meal 4
	Manipulation	Dipping: sauces detected off the plate, dip depth measured with a fallback to the plate’s height
	Interface	Manual keypoint mode taps points on the full camera image and shows the chosen dip
7	Manipulation	Dips spread across the sauce; skewer target offset 5 mm along the item’s major axis; the pose above the plate recalibrated
8	Safety	Default collision threshold, used wherever a skill sets none of its own, raised from 15 to 20 Nm, after a false stop when she retried a bite acquisition at meal 7
9	Perception	Skewer depth measured over a wider patch, with a fallback to the plate’s height; plate-attachment detections larger than the attachment rejected
	Manipulation	The gripper closes before pushing the microwave door
	Interface	The button can also be read on the robot’s computer, as a fallback to the iPad
	Interface	Continuous dictation on every text field
11	Study procedure	The researcher is prompted to position the recording camera before each navigation and plate pickup
12	Perception	Head-nod detection for finishing a sip or a wipe made more robust
	Manipulation	Drinking made functional: the mug is picked from and stowed in a fixed holder on the base, with its transfer distance recalibrated
	Interface	<i>During the meal:</i> a task-selection bug, introduced that morning, fixed
13	Manipulation	Plate-holder pick and place poses recalibrated
14	Navigation	The IMU’s messages re-stamped with the time each is received; before this, the odometry filter could discard the gyroscope and estimate rotation from the wheels alone, spoiling parking
15	Perception	<i>During the meal:</i> the object-detection prompts for two food items corrected
16	Personalization	<i>During the meal:</i> setting labels reduced to personal, social, and television, dropping which side a companion sat on; her stored history relabeled before meal 17
17	Perception	The pose from which the plate is imaged recalibrated
18	Perception	Object-detection prompts tuned automatically from the plate image
	Safety	A takeover can end the wait for a transfer trigger
19	Perception	<i>During the meal:</i> the object-detection prompts for two food items corrected
20	Perception	The robot waits 10 s instead of 5 s for the camera to settle before imaging the plate
21	Perception	Automatic prompt tuning turned off in favor of fixed prompts for each meal’s foods
	Interface	The button is read only on the robot’s computer, after it misfired at meal 20
22	Perception	<i>During the meal:</i> an object-detection prompt added for a sandwich
2–22	Perception	Before most meals, object-detection prompts and meal labels added for that meal’s foods
4, 9, 10	Hardware	<i>During the meal:</i> a gripper fingertip repaired (meal 4), replaced after breaking (meal 9) and retightened (meal 10)
12	Hardware	A fixed holder for the mug added to the base

I TELEOPERATION ACCOUNTING

Base-driving episodes are recovered from `/cmd_vel_teleop` in the deployment rosbags, the only record that survives every route to the base controls, including a detour out of an arm session to reposition an already-parked robot. We separate the researcher’s Xbox controller from the CR’s web interface by publisher identity, and exclude it: it is used to place the base at setup and teardown,

and accounts for more base driving than the CR does. Of the 66 logged arm sessions, 51 command no motion at all — the CR uses the teleoperation screen as an escape hatch to skip past a skill, which on meal 3 she described as only wanting “to skip driving” — and a further 8 jog the arm without the arm being stuck: pressing arm-control is also how she stops the robot to reach the *Move the Base* button that follows *Done*, how a researcher pauses it to adjust



Figure G1: All 22 meals of the deployment, in order, each plate as served before the first bite and cropped to its rim. Meals were chosen by the CR and vary in piece size, count, and spacing, from dry and crumbly (hash brown, pancake) to sauced and slippery (orange chicken); dips are named in italics. Each plate’s disc carries the meal number and is filled with the color of that meal’s setting (Table E1; left panel of Fig. 3 of the main paper). Six crops contain a small region of synthesized bare plate where the plate ran past the camera frame or was occluded by the fork; food is never altered.

hardware, and once how she nudged an idle arm between bites. We count the remaining 7, all rescues of an arm that had hit a joint limit. Bag coverage of the meal windows is 92–100% per day. Both series are shown in the left panel of Fig. 3 of the main paper.

J CORRECTION LEDGER

Table J1 lists every correction the CR made over the deployment, in the order she made them. Corrections reach the system through two channels. Each predicted dimension appears on its own preference page where it takes effect, and she overrides the ones she disagrees with; these appear as *Ask* rows. During the meal, she can open the settings page and change a dimension directly; these appear as *Set* rows. Plate colors she re-picked on the color-correction page appear as *Col* rows; they are the 19 plate-color corrections of the main paper, counted apart from the 57 preference corrections. All three are logged with the value the system held immediately before the override, so each row records what was predicted and what she replaced it with.

K MEASURES

Daily. After every meal, the CR rated 11 items on the iPad, one per screen, on a 7-point scale anchored at *Very Low* (1) and *Very High* (7); each is shaded in the bottom-left panel of Fig. 3 of the main paper so that green is the favorable end. Five are NASA-TLX workload items [5] and six concern her experience of the robot; Table K1 gives the wording of each. One open question followed:

“What, if anything, did you learn about the robot today, or how did your interaction with it change?” Her answers are reproduced in full in Table E2 and quoted in the main paper.

End of study. After the final meal she completed the following, each on a 5-point agreement scale from *strongly disagree* (1) to *strongly agree* (5): the five items of the *Technology Acceptance Model* [4]; the 10 standard items of the *System Usability Scale* [1]; two items on *Control and Independence*, each administered twice, once for the robot and once for her human caregiver, to enable direct comparison; and two items on *Co-Adaptation*. Table K1 gives the wording of all but the System Usability Scale items.

L HUMAN ADAPTATION: PROTOCOL AND ROBUSTNESS

Recovering her triggering aperture. Across the 15 meals with mouth-open triggering active, we re-ran the deployed detector offline over the logged eye-in-hand video and recovered the peak mouth aperture ratio (MAR) at which each of 261 bite requests fired; it recovers an opening for 259 of them. The per-meal medians sit at two levels. Neither block trends internally (meals 1 to 15, $\rho = -0.32$, $p = 0.41$; meals 16 to 22, $\rho = +0.09$, $p = 0.87$), and they separate perfectly at the transition: the lowest median before it (0.674) exceeds the highest after it (0.669). Sweeping every candidate split and maximizing the between-group sum of squares locates the changepoint at meal 16, and a permutation test on that maximum, which corrects for searching all splits, rejects

Table H1: The preference bundle the system held at the end of every meal. One column per meal, headed by the plate as served and that meal's setting; one row per preference dimension. **Shaded bold** cells are dimensions the CR corrected during that meal; every other cell is a value she accepted as predicted, in black where it changed from the meal before (and at meal 1) and in gray where it did not. The continuous dimensions (plate-attachment color at each pickup location, and the per-location parking offsets) and the free-text bite-ordering dimension are omitted. Setting colors are those of the left panel of Fig. 3 of the main paper and of Fig. G1; abbreviations are keyed below the matrix.

PLATE	chicken nuggets	orange chicken	teriyaki chicken	chicken kebab	strawberry + whipped cream	hash brown, sausage	chicken nuggets + ketchup	orange chicken	teriyaki chicken	hash brown	hash brown, sausage	popcorn chicken, hash brown + ranch	orange chicken	chicken kebab	pancake, sausage + maple syrup	steak, mozzarella sticks	chicken popcorn, potato wedges	PB&J sandwich	chicken kebab, pineapple	kielbasa, carrots	watermelon, cantaloupe, strawberry	PB&J sandwich				
SETTING	TV	TV	PERSONAL	SOCIAL	TV	PERSONAL	SOCIAL	TV	SOCIAL	SOCIAL	TV	PERSONAL	SOCIAL	TV	PERSONAL	SOCIAL	SOCIAL	PERSONAL	TV	SOCIAL	PERSONAL	SOCIAL				
MEAL	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22				
Checks with her																										
Confirm navigation arrival	15 s	15 s	skip	15 s	skip	skip	15 s	skip	skip	skip	skip	skip	skip	15 s	skip	skip	skip	skip	skip	skip	skip	skip				
Confirm manipulation	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	skip	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s				
Show bite-selection page	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s	15 s				
Confirm feeding pickup	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip	skip				
Next-task countdown	30 s	30 s	15 s	wait	30 s	15 s	wait	30 s	wait	wait	30 s	15 s	wait	30 s	15 s	60 s	60 s	15 s	30 s	wait	15 s	30 s				
Signals																										
Robot's ready-to-feed cue	SP	SP	LED	LED	LED	LED	LED	LED	LED	LED	LED	SP	LED	LED	SP	LED	LED	SP	LED	LED	SP	SP				
Completion signal	SP	SP	LED	LED	LED	LED	LED	LED	LED	LED	LED	LED	LED	LED	SP	LED	LED	SP	LED	LED	SP	SP				
Bite trigger	OM	OM	OM	BT	OM	OM	AP	OM	BT	BT	OM	OM	BT	OM	OM	OM	OM	OM	OM	BT	OM	OM				
Drink trigger	OM	OM	OM	BT	OM	OM	BT	OM	BT	BT	OM	OM	BT	OM	OM	BT	OM	OM	OM	BT	OM	OM				
Wipe trigger	OM	OM	OM	BT	OM	OM	BT	OM	BT	BT	OM	OM	BT	BT	BT	BT	BT	BT	BT	BT	BT	BT				
Done detection (bite)	AP	PC	AP	PC	AP	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC	PC				
Done detection (drink)	PC	PC	PC	BT	PC	PC	BT	PC	BT	BT	PC	BT	BT	BT	BT	BT	BT	BT	BT	BT	BT	BT				
Done detection (wipe)	PC	AP	PC	BT	PC	PC	BT	PC	BT	BT	PC	BT	BT	BT	BT	BT	BT	BT	BT	BT	BT	BT				
Fit to her abilities																										
Skewering axis	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥	⊥					
Transfer mode	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out	out				
Outside-mouth distance	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near	near				
Bite																										
Bite dipping	none	none	none	none	more	none	more	none	none	none	none	more	none	none	less	none	none	none	none	none	none	none				
Motion																										
Robot speed	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast	fast				
Retract between bites	no	no	no	yes	no	no	yes	no	yes	yes	no	no	yes	no	no	no	no	no	no	yes	no	no				
Heating																										
Microwave time	30 s	30 s	30 s	30 s	off	30 s	30 s	30 s	30 s	30 s	30 s	30 s	30 s	30 s	30 s	off	30 s	off	30 s	30 s	off	off				
CELLS	BT	corrected during that meal					BT	accepted, changed from the meal before					BT	accepted, same as the meal before												
SETTING	Watching TV	Alone	With a friend			Music on	Video call			iPad games																
KEY	15 s, 30 s, 60 s	countdown (confirm and autocontinue)					off	no microwave					⊥	perpendicular to major axis									out	outside-mouth transfer		
	SP	speech	OM	open mouth	BT	button	AP	proceed automatically after a pause					PC	perception	wait	wait for me										

the no-change null ($p = 7.5 \times 10^{-4}$). The fitted levels are 0.796 and 0.592 MAR (95% CIs 0.727–0.864 and 0.536–0.648), and only 3% of the 103 post-transition bites exceed the pre-transition median.

Labeling meals 22 and 9. The trigger log cannot say whether the lower level still separates a request from the rest of a meal, because defining a request as whatever fired sensors the distribution at 0.45 by construction. An author therefore labeled every second of meal 22's 14 detector-armed waits as talking, chewing,

smiling, requesting a bite, or at rest, working from the eye-in-hand video; consecutive request seconds are merged, since a request is one continuous opening. Across 110 seconds of everyday behavior, not one reaches 0.45; the closest approach is a single talking second at 0.425. Bite requests sit at a median of 0.502, and 13 of 16 clear the threshold (Mann-Whitney $p = 6.4 \times 10^{-11}$). She is not opening wide to be safe: her requests clear the threshold by a median of 0.05 MAR, the widest aperture she produces anywhere in the meal

Table J1: Every correction the CR made, by meal. Ask corrections are overrides on a dimension’s preference page, where it takes effect; Set corrections are mid-meal changes from the settings page; Col corrections are plate-attachment colors she re-picked on the color-correction page. Predicted is the value the system held for that dimension immediately before the override. A dot marks each row’s channel and a swatch each logged plate color (key below the table).

MEAL	DIMENSION	PREDICTED	CORRECTED TO	MEAL	DIMENSION	PREDICTED	CORRECTED TO
1	Robot speed	medium	fast	7	Bite trigger	button	proceed automatically after a pause
	Plate color (fridge)	12, 223, 169 (0.10)	9, 189, 189 (0.10)		Bite dipping	less	more
	Microwave time	1 min	30 s	Plate color (dining table)	7, 146, 155 (0.18)	10, 180, 207 (0.18)	
	Plate color (microwave)	12, 223, 169 (0.10)	10, 201, 240 (0.24)	8	Confirm navigation arrival	countdown 15 s	skip
	Outside-mouth distance	medium	near	9	Confirm navigation arrival	countdown 15 s	skip
	Retract between bites	yes	no	Plate color (dining table)	10, 180, 207 (0.18)	13, 206, 161 (0.18)	
	Confirm feeding pickup	countdown 15 s	skip	10	Plate color (dining table)	13, 206, 161 (0.18)	16, 136, 253 (0.15)
	Next-task countdown	countdown 15 s	countdown 30 s	11	Plate color (fridge)	8, 170, 160 (0.17)	10, 207, 123 (0.17)
	Done detection (bite)	perception	proceed automatically after a pause	12	Bite ordering	free text ^a	free text ^a
	Plate color (table)	10, 195, 195 (0.12)	3, 99, 129 (0.12)	Done detection (drink)	perception	button	
2	Plate color (fridge)	9, 189, 189 (0.10)	8, 160, 147 (0.19)	Robot's ready-to-feed cue	LED	speech	
	Plate color (microwave)	9, 175, 200 (0.24)	12, 248, 186 (0.24)	Plate color (movable table)	7, 172, 175 (0.20)	13, 216, 255 (0.20)	
	Done detection (wipe)	perception	proceed automatically after a pause	13	Confirm manipulation	countdown 15 s	skip
	Done detection (bite)	proceed automatically after a pause	perception	14	Confirm navigation arrival	skip	countdown 15 s
	Plate color (table)	5, 120, 140 (0.15)	7, 172, 175 (0.24)	Wipe trigger	open mouth	button	
	Done detection (bite)	proceed automatically after a pause	perception	Done detection (drink)	perception	button	
3	Confirm manipulation	countdown 15 s	skip	15	Robot's ready-to-feed cue	LED	speech
	Confirm manipulation	skip	countdown 15 s		Wipe trigger	open mouth	button
	Confirm navigation arrival	countdown 15 s	skip	Done detection (drink)	perception	button	
	Robot's ready-to-feed cue	speech	LED	Bite dipping	more	less	
	Done detection (bite)	perception	proceed automatically after a pause	Plate color (movable table)	7, 172, 175 (0.20)	4, 52, 118 (0.30)	
	Next-task countdown	countdown 30 s	countdown 15 s	16	Microwave time	30 s	no microwave
	Plate color (table)	5, 135, 150 (0.18)	7, 146, 155 (0.18)	Bite trigger	button	open mouth	
	Plate color (table)	5, 135, 150 (0.18)	7, 146, 155 (0.18)	Drink trigger	open mouth	button	
4	Robot speed	medium	fast	Retract between bites	yes	no	
	Robot's ready-to-feed cue	speech	LED	Next-task countdown	countdown 30 s	countdown 60 s	
	Bite trigger	open mouth	button	Plate color (dining table)	13, 180, 205 (0.18)	11, 200, 140 (0.18)	
	Retract between bites	no	yes	17	Bite trigger	button	open mouth
	Confirm feeding pickup	countdown 15 s	skip		Drink trigger	button	open mouth
	Next-task countdown	countdown 30 s	wait for me	Wipe trigger	open mouth	button	
	Done detection (bite)	proceed automatically after a pause	perception	Next-task countdown	countdown 30 s	countdown 60 s	
	5	Confirm navigation arrival	countdown 15 s	skip	18	Completion signal	LED
Plate color (fridge)		8, 175, 168 (0.15)	10, 200, 129 (0.15)	Wipe trigger		open mouth	button
Robot's ready-to-feed cue		speech	LED	Plate color (movable table)	4, 52, 118 (0.30)	13, 219, 238 (0.30)	
Confirm manipulation		countdown 15 s	skip	19	no corrections		
Confirm manipulation		skip	countdown 15 s	20	Bite trigger	open mouth	button
Plate color (movable table)		7, 150, 158 (0.18)	11, 191, 254 (0.18)	21	Plate color (movable table)	7, 172, 175 (0.20)	10, 181, 192 (0.09)
6	Done detection (bite)	proceed automatically after a pause	perception	22	Robot's ready-to-feed cue	LED	speech
	Bite trigger	open mouth	button				
	Bite trigger	button	open mouth				
	Plate color (movable table)	8, 170, 190 (0.18)	3, 89, 97 (0.17)				

^a Meal 12, bite ordering. Predicted: “Dip each chicken popcorn and hash brown in ranch dressing before feeding.” Corrected to: “I would like to eat each chicken popcorn dipped in ranch dressing. I would like to eat...”

● Ask (pre-meal selection) ● Set (mid-meal update) ● Col (plate color) Plate colors: swatch, then hue 0–179, saturation and value 0–255 (matching range).

is 0.99, and her median aperture while actually taking a bite is 0.756. Meal 9, a social meal in which she used the button, was labeled the same way as a contrast in which the detector was never armed: 79 seconds of talking with a median of 0.348 and a maximum of 0.620, of which 12 (15%) cross 0.45, against none of the 31 talking seconds in meal 22 (Fisher $p = 0.018$; Mann–Whitney on peak MAR $p = 2.3 \times 10^{-6}$).

Three caveats. The labeler saw the aperture trace alongside the video, though not the 0.45 line, so the labels are not blind to the signal; that they cut across the threshold in both directions — three requests below it (0.352, 0.429, 0.437) and three non-requests above it, a whistle at 0.464, a laugh at 0.507 and a head-pose artifact at 0.528 — indicates they track intent rather than magnitude. In meal 9, 27 of 106 talking seconds carry a note that she had

Table K1: The survey items. Items are given as the CR saw them on her iPad, one per screen. The daily items were rated on a 7-point scale from *Very Low* (1) to *Very High* (7), and the end-of-study items on a 5-point agreement scale from *strongly disagree* (1) to *strongly agree* (5). The 10 standard items of the System Usability Scale are not repeated here.

ITEM	QUESTION
After every meal: NASA-TLX workload (5)	
Mental demand	How mentally demanding was using the meal-assistance system?
Physical demand	How physically demanding was using the meal-assistance system?
Temporal demand	How much time pressure did you feel when using the meal-assistance system?
Effort	How hard did you have to work to use the meal-assistance system effectively?
Frustration	How frustrated were you while using the meal-assistance system?
After every meal: her experience of the robot (6)	
Performance	How successful were you in using the meal-assistance system to achieve your goals?
Trust	How much did you trust the robot to do the right thing during today's meal?
Safety	How safe did you feel during today's meal?
Predictability	How well could you predict what the robot was going to do during today's meal?
Robot adaptation	How much do you feel the robot has learned about your preferences?
Satisfaction	Overall, how satisfied were you with today's meal?
End of study: Technology Acceptance Model (5)	
Perceived usefulness	Using this meal-assistance system will make me more independent in eating.
Perceived ease of use	This meal-assistance system is easy to use.
Attitude toward using	Using the meal-assistance system for improving my independence is a good idea.
Behavioral intention to use	Assuming I have access to this meal-assistance system, I predict that I would use it in my daily life.
Perceived enjoyment	I find using this meal-assistance system to be enjoyable.
End of study: Control and Independence, each asked for her caregiver and for the robot (2)	
Control	I feel in control of my feeding experience when assisted by my caregiver / the robot.
Independence	I feel a sense of independence when I receive assistance from my caregiver / the robot.
End of study: Co-Adaptation (2)	
Robot's side	The robot got better at understanding my preferences over the course of the study.
Her side	I got better at working with the robot over the course of the study.

turned toward the researcher or downward; we exclude them, because the inter-ocular distance confirms the turn (76 vs 107 px, $p = 2.7 \times 10^{-11}$) and MAR is unreliable off-axis. Including them would raise the crossing rate to 30%. Finally, this is one meal per condition, and they differ in more than the detector: meal 9 is early and has a friend present, meal 22 is late and on a video call. A second button meal (meal 20) shows the same shift in level but only 1 of 47 talking seconds above threshold, so the crossing rate is not a stable property of the unarmed condition.

Time to take the bite. Across 273 bites in the 20 meals with usable timing — five run past the 3 s panel axis in Fig. 4 of the main paper and are drawn at its cap, the longest at 5.6 s — the same changepoint procedure places a step before meal 4 ($p = 2.1 \times 10^{-3}$). The per-meal medians separate perfectly there (the lowest before it, 1.66 s; the highest after it, 0.99 s); the fitted levels fall from 1.92 s to 0.56 s, a factor of 3.4, and the later block does not trend internally ($\rho = -0.02$, $p = 0.95$). Read instead as a gradual decline across the meals, the effect is not significant ($\rho = -0.40$, $p = 0.084$ over the 20 meal medians); the data support an abrupt drop within the first week followed by 16 flat meals. The step survives aggregation to the meal level (Mann-Whitney on per-meal medians, $p = 0.002$), so it is not an artifact of treating the 273 bites as statistically independent observations.

M ATTENTION MEASUREMENT

We recover head pose per frame with a face-mesh regressor and solvePnP, and report the angle between her head direction and

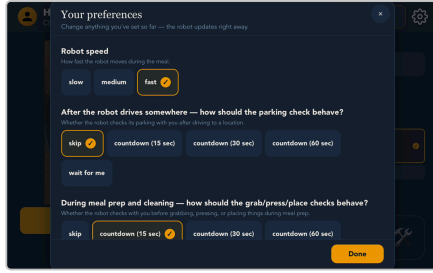
the fork; the eye-in-hand camera sits directly behind the fork, so a small angle means her head is aimed at the robot. The measure covers 357 transfers and 1.45×10^5 frames; meal 14 has no bite-transfer recording and is absent from the per-meal panel. Two windows are compared: the bite request itself (± 1 s) and the rest of the transfer (> 3 s from the request).

Away from the request, her head turns further from the robot as the deployment proceeds ($\rho = +0.74$, $p = 0.011$); the permutation test is stratified by activity so that the late clustering of video-call meals cannot produce the effect. At the request, it does not ($\rho = +0.24$, $p = 0.28$). Across activity types, head direction away from the request varies strongly (Kruskal-Wallis $p = 0.006$, from 14.4° on meals with nothing else to 28.8° on video calls) while at the request it does not ($p = 0.46$, every meal within 10.7 – 19.6°). The activities she chose grew more demanding over the month ($\rho = +0.53$, $p = 0.011$ against an ordering assigned from the labels alone; the five most demanding meals all fall in the last third), and her attention tracks what each one asks of her ($\rho = +0.79$, $p = 1.7 \times 10^{-5}$). The direction of causation is not established by a single-participant observational design.

N INTERFACE SCREENSHOTS

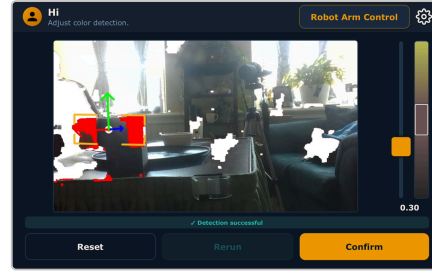
Figures N1 and N2 show the pages of the web interface described in Appendix A, as recorded on the CR's iPad.

PREFERENCE PAGES

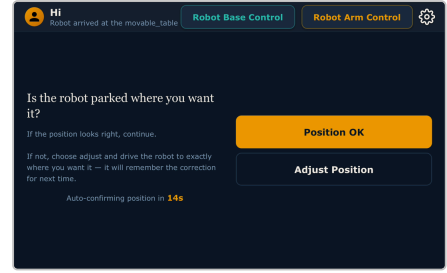


(a) Settings overlay (meal 15)

CONFIRMATION PAGES

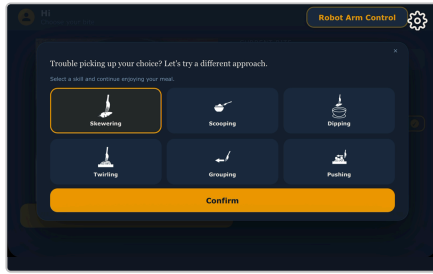


(b) Color correction (meal 15)

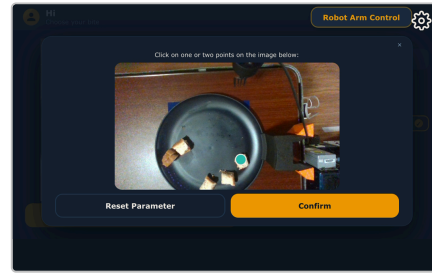


(c) Parking (meal 14)

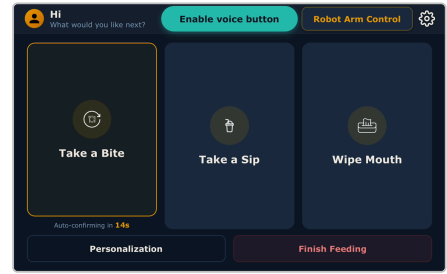
FEEDING PAGES



(d) Manual keypoints: skill (meal 18)

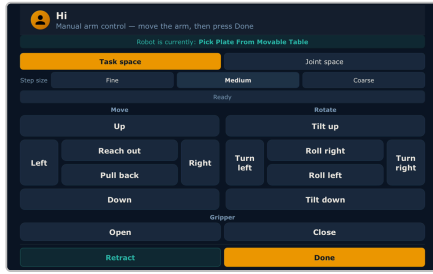


(e) Manual keypoints: points (meal 18)

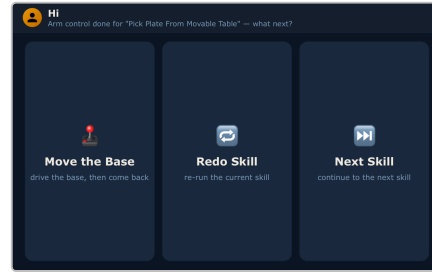


(f) Next task (meal 15)

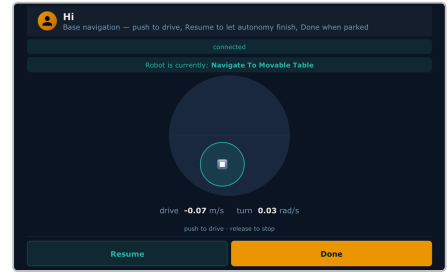
TELEOPERATION PAGES



(g) Arm teleoperation (meal 15)



(h) After teleoperation (meal 15)



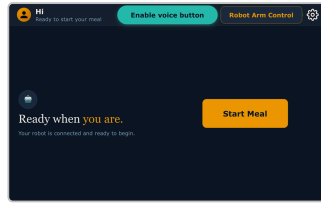
(i) Base teleoperation (meal 14)

Figure N1: Pages that meal 20 did not reach, from the CR's iPad recordings of meals 14, 15, and 18. In (b) she has tapped the plate attachment; the pixels matching its color are shown in red and the slider sets the matched range. (d, e) show the manual keypoint mode, opened from *Trouble picking this up?* on the bite-selection page: she chooses a skill and taps a point on the food. In (i) she drives the base while the robot navigates to the movable table. Background photographs in (b) are blurred. The pages are shown at a larger scale than in Fig. N2.

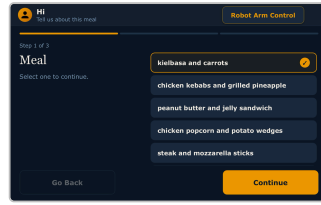
REFERENCES

- [1] John Brooke. 1996. SUS: A quick and dirty usability scale. In *Usability Evaluation in Industry*, Patrick W. Jordan, Bruce Thomas, Bernard A. Weerdmeester, and Ian L. McClelland (Eds.). Taylor & Francis, London, 189–194.
- [2] C. K. Chow and C. N. Liu. 1968. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory* 14, 3 (1968), 462–467.
- [3] Christopher Crick, Graylin Jay, Sarah Osentoski, Benjamin Pitzer, and Odest Chadwicke Jenkins. 2017. Rosbridge: ROS for non-ROS users. In *Robotics Research: The 15th International Symposium ISRR*. Springer, 493–504.
- [4] Fred D. Davis, Richard P. Bagozzi, and Paul R. Warshaw. 1989. User acceptance of computer technology: A comparison of two theoretical models. *Management Science* 35, 8 (1989), 982–1003.
- [5] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Human Mental Workload*, Peter A. Hancock and Najmedin Meshkati (Eds.). Advances in Psychology, Vol. 52. North-Holland, Amsterdam, 139–183.
- [6] Rajat Kumar Jenamani, Tom Silver, Ben Dodson, Shiqin Tong, Anthony Song, Yuting Yang, Ziang Liu, Benjamin Howe, Aimee Whitneck, and Tapomayukh Bhattacharjee. 2025. FEAST: A flexible mealtime-assistance system towards in-the-wild personalization. In *Robotics: Science and Systems (RSS)*.
- [7] Amal Nanavati, Ethan K. Gordon, Taylor A. Kessler Faulkner, Yuxin (Ray) Song, Johnathan Ko, Tyler Schrenk, Vy Nguyen, Bernie Hao Zhu, Haya Bolotski, Atharva Kashyap, Sriram Kutty, Raida Karim, Liander Rainbolt, Rosario Scalise, Hanjun Song, Ramon Qu, Maya Cakmak, and Siddhartha S. Srinivasa. 2025. Lessons learned from designing and evaluating a robot-assisted feeding system for out-of-lab use. In *Proceedings of the 2025 ACM/IEEE International Conference on Human–Robot Interaction*.
- [8] Tom Silver, Rajat Kumar Jenamani, Ziang Liu, Ben Dodson, and Tapomayukh Bhattacharjee. 2025. Coloring between the lines: Personalization in the null space of planning constraints. arXiv:2505.15503.
- [9] Evan You. 2025. Vue.js. <https://vuejs.org> Accessed January 1, 2025.

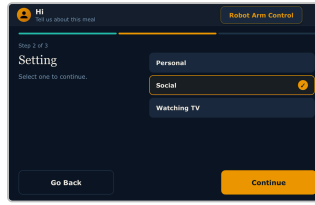
START OF A MEAL



(a) Home

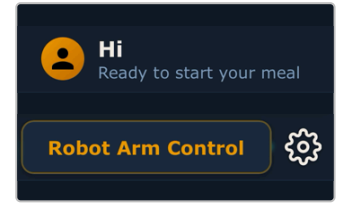


(b) Meal context: meal



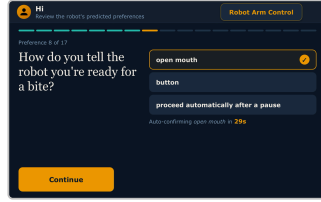
(c) Meal context: setting

SHARED HEADER

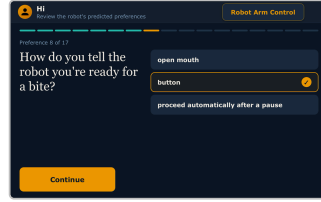


Header of (a), enlarged

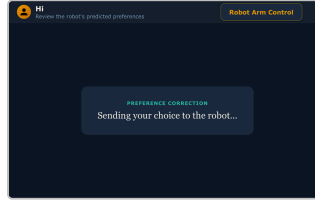
PREFERENCE PAGES



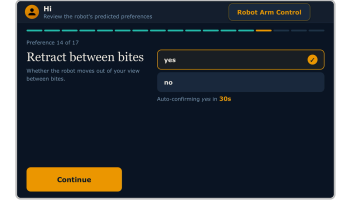
(d) Predicted value



(e) Her correction

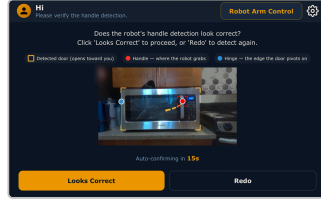


(f) Re-predicting

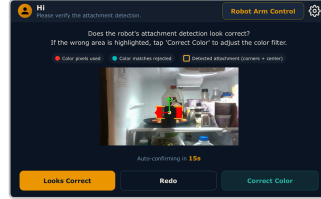


(g) Re-predicted value

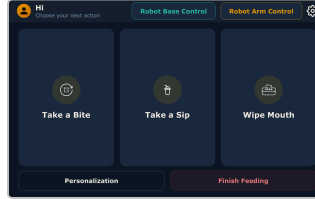
CONFIRMATION PAGES



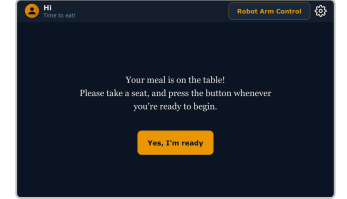
(h) Detection: handle



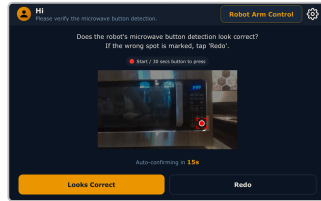
(i) Detection: plate attachment



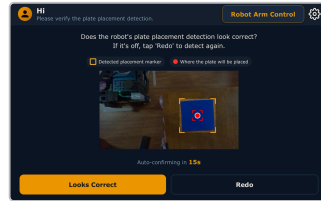
(n) Task selection



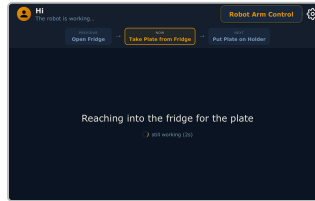
(o) Ready to eat (confirmation page)



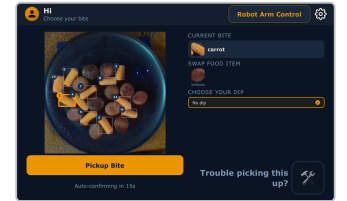
(j) Detection: microwave button



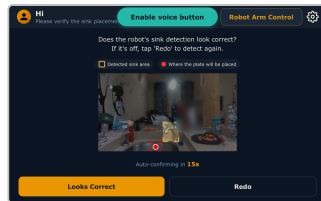
(k) Detection: table placement



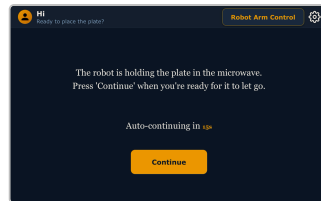
(p) Executing



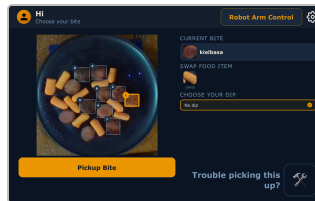
(q) Bite selection: proposed



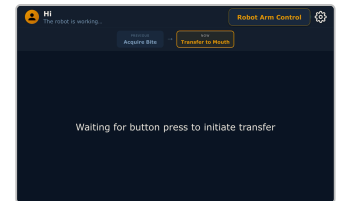
(l) Detection: sink placement



(m) Plate release

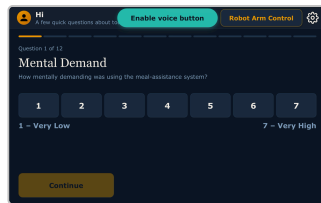


(r) Bite selection: swapped

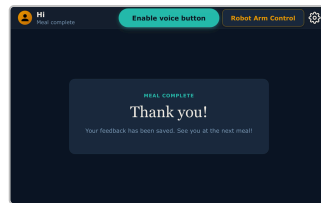


(s) Waiting for her trigger

END OF A MEAL



(t) End-of-meal survey



(u) Closing

Figure N2: The web interface at meal 20, as recorded on the CR's iPad. The header, enlarged at top right, carries the status line and the *Robot Arm Control* button on every page, and the settings gear on most; her name is erased from it. (d–g) show one correction and its re-prediction: the robot predicted open mouth as her bite trigger (d), she chose the button (e), and the robot re-predicted the open dimensions (f), revising retraction between bites (g) and the next-task countdown, to waiting for her. (h–l) are the five detections the detection page confirms. (q, r) show her swapping the proposed bite to another food item.